

Unveiling Ancient Symmetry: Islamic Girih Tiles and the Mathematics of Aperiodicity

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This essay explores the intersection of art and mathematics through the lens of Islamic architectural design, specifically focusing on the presence of quasiperiodic patterns. While seemingly random, these patterns possess order and structure without exact repetition, a concept that challenges traditional notions of symmetry. The essay examines the role of girih tiles, a set of five geometric shapes documented in the 15th-century Topkapi Scroll, used by Islamic artisans to create intricate, non-repeating designs. These tilings, notably found in structures like the 15th-century Darb-i Imam Shrine, bear striking resemblance to Penrose tilings, mathematical constructs developed in the 1970s that exhibit five-fold rotational symmetry— a characteristic previously deemed impossible in crystallography. The essay highlights the debate surrounding the intentionality of Islamic artisans in creating quasiperiodic patterns. While some scholars argue that these patterns were unintentional byproducts of the girih tile system, others propose that the artisans possessed a sophisticated understanding of geometric principles which prominently features in Penrose tilings. The essay concludes by emphasizing the remarkable convergence of art and mathematics in Islamic architecture. Whether consciously intended or not, the presence of quasiperiodic patterns in these ancient structures showcases the depth of geometric knowledge possessed by Islamic artisans and their remarkable ability to translate complex mathematical concepts into visually stunning works of art.

1 ISLAMIC GIRIH TILES: PRECURSORS TO PENROSE TILINGS?

In 1974, Roger Penrose unveiled a revolutionary concept now known as non-periodic tiling, or Penrose tiling: a five-fold symmetrical pattern that, despite its order, never repeats. This challenged the conventional understanding of tiling and symmetry in the scientific community. Years later, in 2011, Dan Shechtman received a Nobel Prize for his groundbreaking discovery of quasicrystals, a three-dimensional manifestation of Penrose tiling in the atomic structure of certain

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VERSION January 5, 2025



Figure 1: Harvard University Peter Lu, left, and his cousin Christina Tam in Bukhara, Uzbekistan on Feb. 21, 2007

alloys [1]. Shechtman's discovery, initially met with resistance and disbelief from prominent scientists such as Linus Pauling, ultimately led to a paradigm shift in how scientists perceive the very nature of matter [1]. What was once considered a "forbidden symmetry" became a recognized and fascinating phenomenon in the world of crystals [1]. The scientific breakthroughs of Penrose and Shechtman have spurred immense interest in the study of nonperiodic patterns and their implications. However, a surprising discovery in 2005 added another layer of intrigue to this narrative. Peter J. Lu, a graduate student at the time, found evidence of these seemingly modern nonperiodic patterns in ancient mosaics that adorn a 17th-century madrasah in Uzbekistan [2]. This finding pushed back the known existence of these patterns by centuries, predating Penrose's work by approximately 240 years [2]. To appreciate the significance of this discovery, it is crucial to understand the fundamental characteristics of non-periodic tiling. Penrose tilings are aperiodic, which means they lack translational symmetry: sections of the pattern do not repeat identically. However, they exhibit both reflectional symmetry and five-fold rotational symmetry, a property that was considered impossible in traditional crystallography [3]. One well-known example is Penrose's P3 tiling, constructed using two rhombus shapes, which visually demonstrates the lack of repeating clusters within the pattern [3]. The use of five-fold symmetry is key to achieving aperiodicity in Penrose tilings. Two, three, four, and six-fold symmetries would inevitably lead to periodic patterns when extended on an infinite plane [3]. The presence of what seemed to be Penrose tilings in ancient

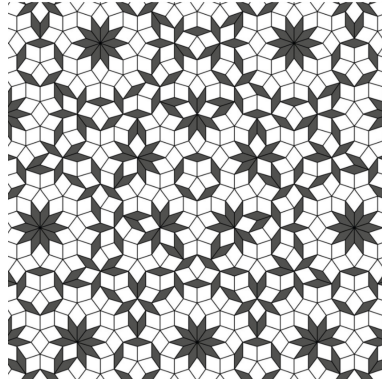


Figure 2: P3 Tiling

Islamic architecture prompted the question: Did Islamic artisans intentionally create these mathematically sophisticated patterns, or was it an unintentional byproduct of their intricate design practices?

2 THE ROLE OF GIRIH TILES AND ISLAMIC ANICONISM

Lu's investigation into Islamic architectural patterns led him to the Topkapi Scroll, a 15th-century architectural guide containing a collection of geometric patterns constructed using five specific shapes: a decagon, pentagon, hexagon, bowtie, and rhombus [2]. These shapes, known as girih tiles, could be arranged to create a vast array of intricate designs. Upon closer examination, Lu realized that these girih tiles could be further subdivided into the kite and dart shapes found in Penrose's P2 tiling [2]. The use of girih tiles, as evidenced by the Topkapi Scroll, offered a practical and efficient method for creating complex patterns. By utilizing a few full-sized templates, artisans could accurately transfer the designs onto plaster surfaces, ensuring precision and minimizing errors that might occur when drafting directly onto walls [2]. The source mentions that the Darb-i Imam shrine, built in 1453, features a near-perfect Penrose tiling using girih tiles, suggesting that Islamic artisans had achieved these patterns centuries before their formal mathematical description [2]. The development of these sophisticated geometric patterns in Islamic art can be partially attributed to the Islamic principle of aniconism, the avoidance of representational imagery in religious art. While not explicitly prohibited in the Quran, the prevailing belief was that depicting living beings constituted a form of competition with God's creative power [4]. This led to

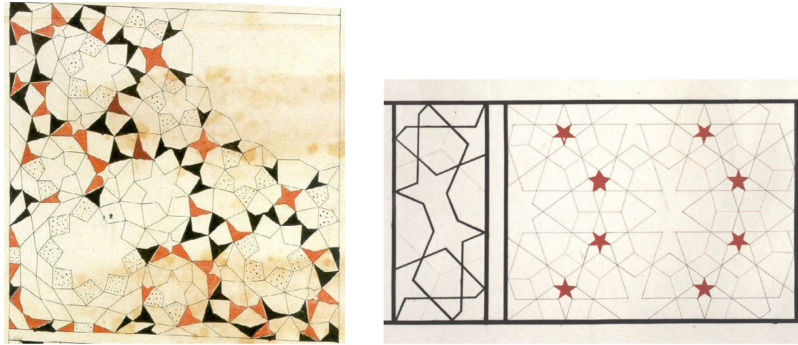


Figure 3: Pages from the Topkapi Scroll

a focus on abstract forms, particularly geometric and floral patterns, to express religious themes and evoke a sense of the divine [5]. Islamic aniconism fostered a rich tradition of geometric exploration, leading to the creation of intricate patterns like those found in the Mosque of Isfahan (also known as the Shah Mosque). Its seemingly simple blue exterior conceals a breathtaking array of infinitely repeating patterns formed by interlocking polygons and ornate calligraphy [4,5]. The mosque, built during the Ilkhanid dynasty, exemplifies the flourishing of Islamic art following the Mongols' conversion to Islam [4]. The Mongol conquest also brought about a thriving paper industry in Iran, leading to the widespread use of architectural scrolls like the Topkapi Scroll to disseminate innovative designs [2].

3 THE DEBATE AND LASTING IMPACT

While the presence of quasiperiodic patterns in Islamic art is undeniable, the question of whether Islamic artisans consciously understood and aimed to create these patterns remains a subject of debate. Scholars like Peter R. Cromwell and Emil Makovicky have questioned whether artisans sought to achieve infinitely expandable quasiperiodic patterns, as there is no direct evidence to support this notion [2]. They argue that the patterns might have emerged as a natural consequence of working with the girih tile system, rather than a deliberate attempt to create aperiodicity. The sources also mention that the original set of five girih tiles has been expanded with time, adding complexity to the discussion about their historical use [2]. However, regardless of whether these patterns were intentionally designed or accidentally discovered, their existence centuries before the

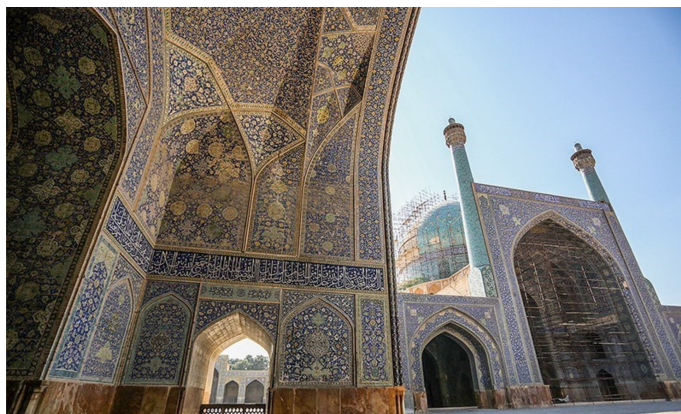


Figure 4: The Mosque of Isfahan

formalization of quasiperiodic mathematics in the West testifies to the remarkable ingenuity and sophisticated geometrical knowledge of Islamic artisans [2]. Their work showcases a profound grasp of geometric principles, including symmetry and proportion, and explores the potential for creating aperiodic tilings, even if these ideas were not explicitly framed in the modern mathematical language. Peter J. Lu's research on Islamic art and quasiperiodic structures has illuminated the mathematical sophistication and artistic ingenuity of these designs, while also inspiring renewed interest in applications of quasicrystals. These materials, such as the quasicrystal icosahedral Yb-Cd [6], possess unique properties that arise from their aperiodic atomic arrangement, making them valuable in various modern technologies, including surgical instruments, LED lights, and non-stick coatings [2].

In conclusion, the Islamic tradition of geometric patterning, driven in part by aniconism, led to the creation of intricate designs that predate and foreshadow the mathematical discovery of Penrose tilings. Although the intentionality behind these patterns remains a matter of scholarly debate, their existence serves as a powerful testament to the advanced geometrical knowledge and artistic innovation present in Islamic art centuries ago. Lu's research, building on the legacy of these ancient artisans, has not only deepened our understanding of Islamic art, but also contributed to the development of modern materials with unique properties, demonstrating the enduring relevance and impact of these timeless patterns.

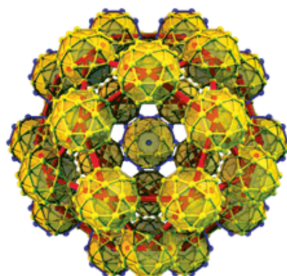


Figure 5: An icosahedral Yb-Cd quasicrystal is composed of an aperiodic ordered arrangement of YbCd clusters (yellow and blue spheres at the vertices of the polyhedra).

REFERENCES

- [1] Adversity, Quasicrystals and a Nobel - the Forbidden Fivefold Symmetry that Was. Andrei Mihai (2022). <https://www.lindau-nobel.org/blog-adversity-quasicrystals-and-a-nobel-the-forbidden-fivefold-symmetry-that-was/>
- [2] Ancient Islamic Penrose Tiles. Julie Rehmeyer (2007). <https://www.sciencenews.org/article/ancient-islamic-penrose-tiles>
- [3] Tessellation, Penrose Tilings and Infinity – TOM ROCKS MATHS. Harriet Wood (2020). <https://tomrocksmaths.com/2020/05/14/tessellation-penrose-tilings-and-infinity/>
- [4] Islamic arts - Mosques, Architecture, Calligraphy - Britannica.pdf. Oleg Grabar and Jacob M. Landau (2024).
- [5] Fivefold Symmetry: A Lesser Symmetry. J. Rosen (1992). https://www.worldscientific.com/doi/abs/10.1142/9789814350807_0002
- [6] Quasicrystal Structure Solved. Mitch Jacoby (2007). <https://cen.acs.org/articles/85/i3/Quasicrystal-Structure-Solved.html>